# Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI


## CONTACT

anika.sharma@example.org
+91 00000 12345
New Delhi, India
github.com/anika-sharma-example
linkedin.com/in/anika-sharma-example
anika-sharma.example.org


## TECHNICAL SKILLS

**Avionics**
MIL-STD-1553 / ARINC 429, Avionics integration

**Scientific instrumentation**
Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems

**Embedded systems**
Embedded C, ESP32 / STM32 microcontrollers

**Computer vision**
Image super-resolution, Object detection

**Drones & UAV systems**
UAV payload integration

**Edge AI**
NVIDIA Jetson deployment

**AI / machine learning**
Deep learning, PyTorch

## EDUCATION

**Ph.D., Electrical and Electronics Engineering**
*Northfield Institute of Science and Technology*
2021 – ongoing
**Thesis:** Lightweight deep networks for real-time enhancement of aerial imagery

**M.Tech., Avionics**
*Coastal Institute of Aeronautical Engineering*
2010
**Grade:** CGPA 8.6/10
**Thesis:** Fault-tolerant data acquisition for airborne test instrumentation

**B.E., Electronics and Communication Engineering**
*Northfield Institute of Science and Technology, Northfield University*
2004
**Grade:** First Class with Distinction

## CERTIFICATIONS & TRAINING

## PROFESSIONAL SUMMARY


Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.


## PROFESSIONAL EXPERIENCE

**Scientist E** Jul 2016 – Present
*Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi*
Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.
**Tools:** Python, PyTorch, MQTT, Node-RED, InfluxDB, Grafana, Docker

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

*Additional charge:* **Workshop In-charge**, Upper Air Division Workshop Apr 2019 – Present
Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

- Supervise technicians and plan preventive maintenance for workshop test equipment.
- Introduced a job-card and spares-inventory system for instrument repairs and calibration work.

*Nodal officer:* **Nodal Officer, IoT and AI/ML Initiatives** Jan 2021 – Present

**Senior Design Engineer** Jun 2004 – Jun 2016
*Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru*
Integration and qualification testing of airborne electronics and UAV payloads.
**Tools:** MIL-STD-1553, ARINC 429, LabVIEW, C, MATLAB

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.
- Integrated mission-computer and display units on MIL-STD-1553 avionics buses and built the automated test rig used for bench validation.
- Planned environmental qualification (MIL-STD-810 vibration, temperature and humidity) for line-replaceable units and documented the test results.

**Graduate Engineer** Aug 2002 – May 2004
*Test Systems Group, Meridian Electronics Limited, Bengaluru*
Built automated test equipment for avionics line-replaceable units.
**Tools:** LabVIEW, C

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.




*Fictional sample profile — not a real person*

**Deep Learning for Edge Devices (40-hour course)**
*Example Academy of Applied Computing*
2022

**Public Procurement and GFR 2017 (5-day programme)**
*Example Institute of Public Administration*
2019

## KEY PROJECTS

**Edge super-resolution pipeline for UAV imagery** *(Ongoing)* Sep 2021 – Present
*Principal investigator, Kestrel Atmospheric Instrumentation Laboratory*
Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.
**Technologies:** NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

**Radiosonde ground-station acceptance test protocol** Jan 2018 – Mar 2019
*Coordinator*
Station acceptance testing varied between sites.

- A common acceptance protocol and checklist adopted for new ground-receiving systems.

**IoT monitoring of laboratory infrastructure** Feb 2020 – Jun 2022
*Project lead*
Instrument rooms lacked continuous environmental and power-quality monitoring.
**Technologies:** ESP32, Environmental sensors, MQTT, Node-RED, InfluxDB, Grafana

- Deployed ESP32 sensor nodes publishing over MQTT to a Node-RED flow, InfluxDB time-series storage and Grafana dashboards with threshold alerts.
- Continuous monitoring with automated alerts replaced manual twice-daily readings.

## PUBLICATIONS

[1] **A. Sharma** and R. Iyer, "Knowledge-distilled lightweight networks for drone image super-resolution," accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, "Benchmarking embedded GPUs for real-time image enhancement," Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, "Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs," Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, "An IoT telemetry framework for upper-air ground stations," Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, "Edge telemetry dashboards for upper-air ground stations," Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, "A low-cost GNSS receiver chain for radiosonde tracking," Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

## MANUSCRIPTS UNDER REVIEW / IN PREPARATION

[1] **A. Sharma**, M. Khan and R. Iyer, "Detection-aware quality assessment of super-resolved drone images," under review at Example Transactions on Geoscience Imaging. *[Under review]*

[2] **A. Sharma**, "Simulation study of balloon-released gliding sondes," in preparation. *[In preparation]*

## AWARDS & HONOURS

**Best Paper Award** 2023
*Example Conference on Instrumentation (ECI 2023)*



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